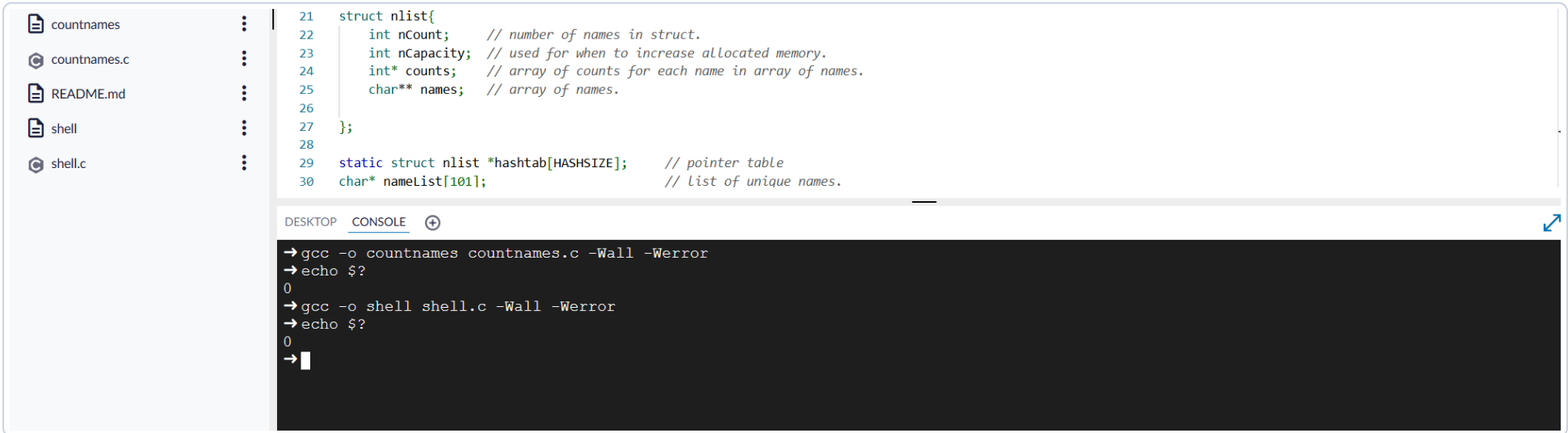


1. Clean compilation

Compiled with `gcc -o countnames countnames.c -Wall -Werror` and `gcc -o shell shell.c -Wall -Werror`; both exit status 0.

Full screenshot: files and console



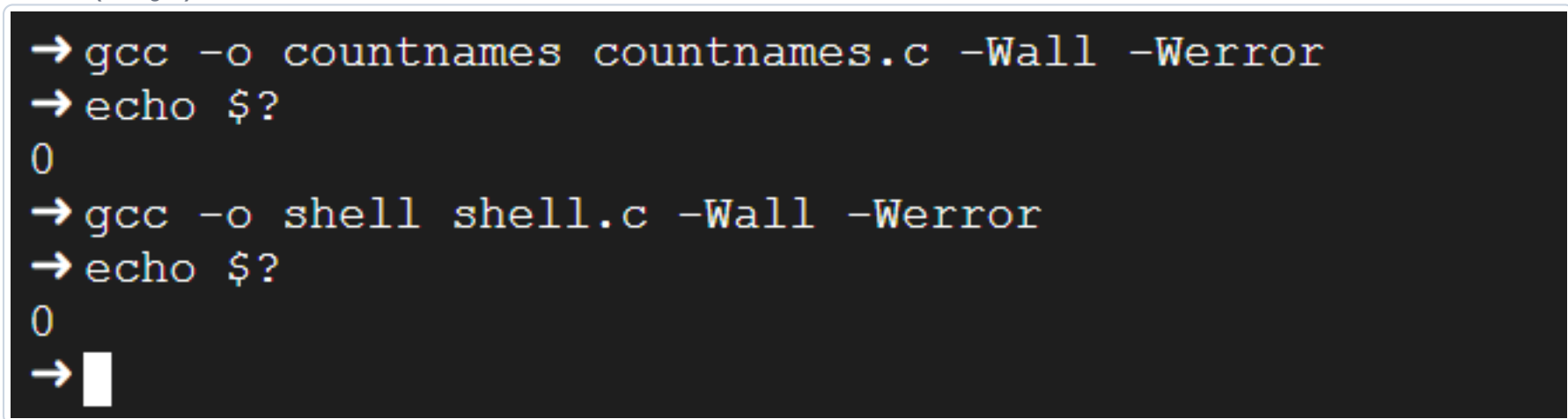
The screenshot shows a code editor interface. On the left, a file explorer lists: `countnames`, `countnames.c`, `README.md`, `shell`, and `shell.c`. The main editor area displays the following C code:

```
21 struct nlist{
22     int nCount;    // number of names in struct.
23     int nCapacity; // used for when to increase allocated memory.
24     int* counts;   // array of counts for each name in array of names.
25     char** names;  // array of names.
26 }
27 };
28
29 static struct nlist *hashtab[HASHSIZE]; // pointer table
30 char* nameList[101];                  // list of unique names.
```

Below the code is a console window with the following output:

```
→ gcc -o countnames countnames.c -Wall -Werror
→ echo $?
0
→ gcc -o shell shell.c -Wall -Werror
→ echo $?
0
→
```

Console (enlarged)



```
→ gcc -o countnames countnames.c -Wall -Werror
→ echo $?
0
→ gcc -o shell shell.c -Wall -Werror
→ echo $?
0
→
```

2. Provided tests: names1.txt and names2.txt

shell forks one child per input file; each child writes <PID>.out (stdout) and <PID>.err (stderr).

Terminal

```
→ ./shell test/names1.txt
Child 492 terminated normally with exit code: 0
→ ./shell test/names2.txt
Child 499 terminated normally with exit code: 0
```

Child 492 - test/names1.txt

492.out (stdout)

492.err ✕ 492.out ✕

```
1 Tom Wu: 3
2
```

492.err (stderr)

492.err ✕ 492.out ✕

```
1 Warning - file test/names1.txt line 3 is empty.
2
```

Child 499 - test/names2.txt

499.out (stdout)

499.err ✕ 499.out ✕

```
1 Jenn Xu: 2
2 Tom Wu: 1
3
```

499.err (stderr)

499.err ✕ 499.out ✕

```
1
```

3. Provided tests: names.txt and namesB.txt

Duplicate counting and multi-word names go to .out; empty-line warnings go to .err.

Terminal

```
→ ./shell test/names.txt
Child 506 terminated normally with exit code: 0
→ ./shell test/namesB.txt
Child 513 terminated normally with exit code: 0
```

Child 506 - test/names.txt

506.out (stdout)

```
506.err x 506.out x
1 Nicky: 1
2 Dave Joe: 2
3 Yuan Cheng Chang: 3
4 John Smith: 1
5
```

506.err (stderr)

```
506.err x 506.out x
1 Warning - file test/names.txt line 2 is empty.
2 Warning - file test/names.txt line 5 is empty.
3
```

Child 513 - test/namesB.txt

513.out (stdout)

```
513.err x 513.out x
1 Nicky: 1
2 Dave Joe: 2
3 Yuan Cheng Chang: 3
4 John Smith: 1
5
```

513.err (stderr)

```
513.err x 513.out x
1 Warning - file test/namesB.txt line 2 is empty.
2 Warning - file test/namesB.txt line 5 is empty.
3
```

4. Large provided tests (overview)

names_long.txt and the four names_long_redundant files; each child's .out and .err are on the next five pages.

Terminal: five runs, one child per input file

```
→ ./shell test/names_long.txt
Child 520 terminated normally with exit code: 0
→ ./shell test/names_long_redundant.txt
Child 527 terminated normally with exit code: 0
→ ./shell test/names_long_redundant1.txt
Child 535 terminated normally with exit code: 0
→ ./shell test/names_long_redundant2.txt
Child 542 terminated normally with exit code: 0
→ ./shell test/names_long_redundant3.txt
Child 549 terminated normally with exit code: 0
→ □
```

Which child ran which file

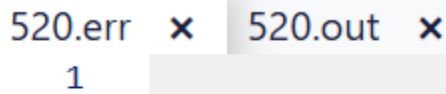
Child	Input file
520	test/names_long.txt
527	test/names_long_redundant.txt
535	test/names_long_redundant1.txt
542	test/names_long_redundant2.txt
549	test/names_long_redundant3.txt

Note: the .out screenshots on the following pages show only the first 39 to 40 lines of each file. These outputs are longer than one screenshot, so the remaining lines are not visible in the images. Each .err file is shown in full.

4. Large provided test: names_long.txt

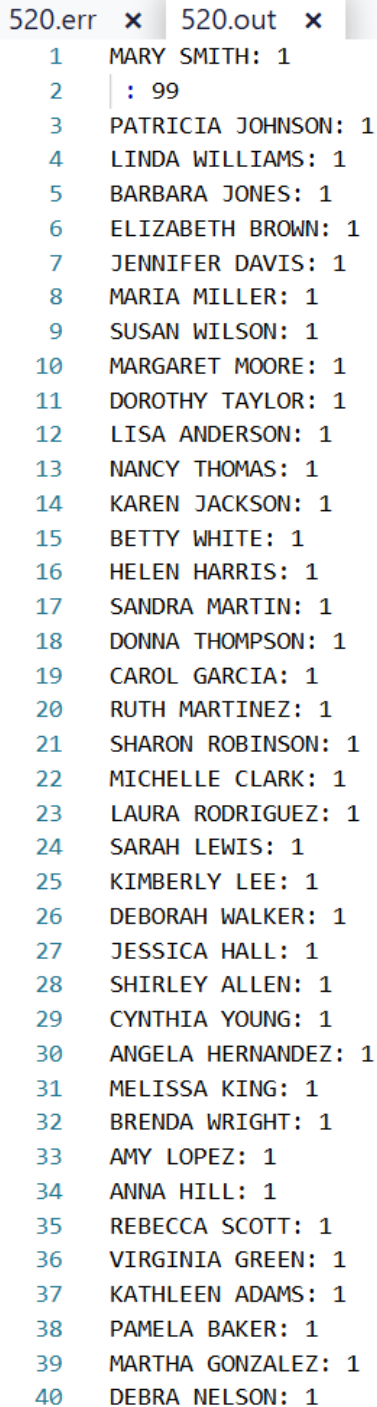
Child 520: ./shell test/names_long.txt. The .out screenshot is truncated (first 40 lines only).

520.err (stderr)



```
520.err x 520.out x
1
```

520.out (stdout)



```
520.err x 520.out x
1 MARY SMITH: 1
2 : 99
3 PATRICIA JOHNSON: 1
4 LINDA WILLIAMS: 1
5 BARBARA JONES: 1
6 ELIZABETH BROWN: 1
7 JENNIFER DAVIS: 1
8 MARIA MILLER: 1
9 SUSAN WILSON: 1
10 MARGARET MOORE: 1
11 DOROTHY TAYLOR: 1
12 LISA ANDERSON: 1
13 NANCY THOMAS: 1
14 KAREN JACKSON: 1
15 BETTY WHITE: 1
16 HELEN HARRIS: 1
17 SANDRA MARTIN: 1
18 DONNA THOMPSON: 1
19 CAROL GARCIA: 1
20 RUTH MARTINEZ: 1
21 SHARON ROBINSON: 1
22 MICHELLE CLARK: 1
23 LAURA RODRIGUEZ: 1
24 SARAH LEWIS: 1
25 KIMBERLY LEE: 1
26 DEBORAH WALKER: 1
27 JESSICA HALL: 1
28 SHIRLEY ALLEN: 1
29 CYNTHIA YOUNG: 1
30 ANGELA HERNANDEZ: 1
31 MELISSA KING: 1
32 BRENDA WRIGHT: 1
33 AMY LOPEZ: 1
34 ANNA HILL: 1
35 REBECCA SCOTT: 1
36 VIRGINIA GREEN: 1
37 KATHLEEN ADAMS: 1
38 PAMELA BAKER: 1
39 MARTHA GONZALEZ: 1
40 DEBRA NELSON: 1
```

Note: this screenshot shows only the first 40 lines of 520.out. The file continues past the bottom of the screenshot, so the remaining lines are not shown.

4. Large provided test: names_long_redundant.txt

Child 527: ./shell test/names_long_redundant.txt. The .out screenshot is truncated (first 39 lines only).

527.err (stderr)

527.err x 527.out x

```
1 Warning - file test/names_long_redundant.txt line 97 is empty.
2
```

527.out (stdout)

527.err x 527.out x

```
1 John Smith: 1
2 Emily Davis: 1
3 Michael Johnson: 1
4 Sophia Lee: 1
5 Daniel Rodriguez: 1
6 Mia Hernandez: 1
7 David Kim: 1
8 Olivia Brown: 1
9 Anthony Martinez: 1
10 Evelyn Jackson: 1
11 Christopher Gonzalez: 1
12 Avery Taylor: 2
13 Andrew Lee: 2
14 Aria Perez: 1
15 Matthew Davis: 1
16 Victoria Wilson: 1
17 William Anderson: 1
18 Madison Green: 1
19 Joshua Wright: 1
20 Isabella Clark: 1
21 Nathan Davis: 2
22 Natalie Adams: 1
23 Ethan Martin: 1
24 Chloe Baker: 1
25 Ryan Lee: 1
26 Samantha Perez: 1
27 Benjamin Wright: 1
28 Aaliyah Garcia: 1
29 Jacob Cooper: 1
30 Elizabeth Martinez: 1
31 Tyler Thompson: 1
32 Hannah Scott: 1
33 Alexander Rodriguez: 1
34 Makayla Turner: 1
35 Nicholas Nguyen: 1
36 Sofia Hernandez: 1
37 Brandon Smith: 1
38 Grace Lee: 1
39 Christian Martinez: 1
```

Note: this screenshot shows only the first 39 lines of 527.out. The file continues past the bottom of the screenshot, so the remaining lines are not shown.

4. Large provided test: names_long_redundant1.txt

Child 535: ./shell test/names_long_redundant1.txt. The .out screenshot is truncated (first 39 lines only).

535.err (stderr)

535.err x 535.out x

```
1 Warning - file test/names_long_redundant1.txt line 2 is empty.
2 Warning - file test/names_long_redundant1.txt line 4 is empty.
3 Warning - file test/names_long_redundant1.txt line 6 is empty.
4 Warning - file test/names_long_redundant1.txt line 8 is empty.
5
```

535.out (stdout)

535.err x 535.out x

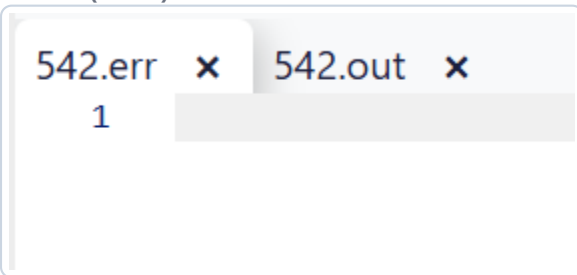
```
1 MARY SMITH: 1
2 PATRICIA JOHNSON: 1
3 LINDA WILLIAMS: 1
4 BARBARA JONES: 1
5 ELIZABETH BROWN: 1
6 JENNIFER DAVIS: 1
7 MARIA MILLER: 1
8 SUSAN WILSON: 1
9 MARGARET MOORE: 1
10 DOROTHY TAYLOR: 1
11 LISA ANDERSON: 1
12 NANCY THOMAS: 1
13 KAREN JACKSON: 1
14 BETTY WHITE: 1
15 HELEN HARRIS: 1
16 SANDRA MARTIN: 1
17 DONNA THOMPSON: 1
18 CAROL GARCIA: 1
19 RUTH MARTINEZ: 1
20 SHARON ROBINSON: 1
21 MICHELLE CLARK: 1
22 LAURA RODRIGUEZ: 1
23 SARAH LEWIS: 1
24 KIMBERLY LEE: 1
25 DEBORAH WALKER: 1
26 JESSICA HALL: 1
27 SHIRLEY ALLEN: 1
28 CYNTHIA YOUNG: 1
29 ANGELA HERNANDEZ: 1
30 MELISSA KING: 1
31 BRENDA WRIGHT: 1
32 AMY LOPEZ: 1
33 ANNA HILL: 1
34 REBECCA SCOTT: 1
35 VIRGINIA GREEN: 1
36 KATHLEEN ADAMS: 1
37 PAMELA BAKER: 1
38 MARTHA GONZALEZ: 1
39 DEBRA NELSON: 1
```

Note: this screenshot shows only the first 39 lines of 535.out. The file continues past the bottom of the screenshot, so the remaining lines are not shown.

4. Large provided test: names_long_redundant2.txt

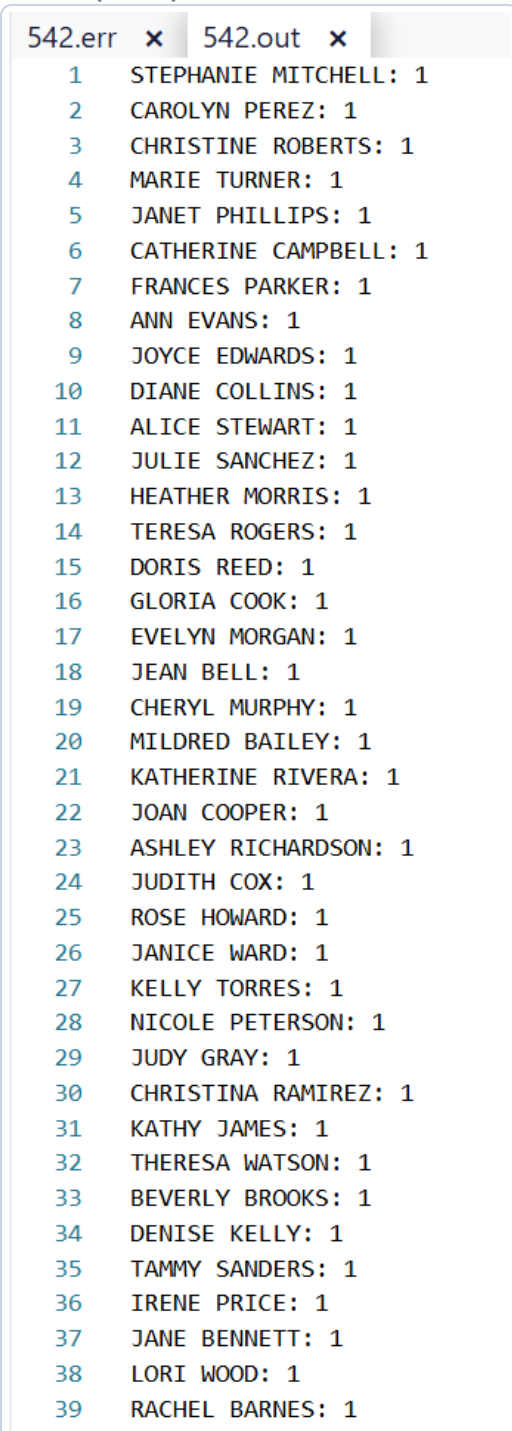
Child 542: ./shell test/names_long_redundant2.txt. The .out screenshot is truncated (first 39 lines only).

542.err (stderr)



```
542.err x 542.out x
1
```

542.out (stdout)



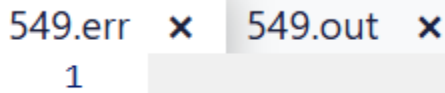
```
542.err x 542.out x
1 STEPHANIE MITCHELL: 1
2 CAROLYN PEREZ: 1
3 CHRISTINE ROBERTS: 1
4 MARIE TURNER: 1
5 JANET PHILLIPS: 1
6 CATHERINE CAMPBELL: 1
7 FRANCES PARKER: 1
8 ANN EVANS: 1
9 JOYCE EDWARDS: 1
10 DIANE COLLINS: 1
11 ALICE STEWART: 1
12 JULIE SANCHEZ: 1
13 HEATHER MORRIS: 1
14 TERESA ROGERS: 1
15 DORIS REED: 1
16 GLORIA COOK: 1
17 EVELYN MORGAN: 1
18 JEAN BELL: 1
19 CHERYL MURPHY: 1
20 MILDRED BAILEY: 1
21 KATHERINE RIVERA: 1
22 JOAN COOPER: 1
23 ASHLEY RICHARDSON: 1
24 JUDITH COX: 1
25 ROSE HOWARD: 1
26 JANICE WARD: 1
27 KELLY TORRES: 1
28 NICOLE PETERSON: 1
29 JUDY GRAY: 1
30 CHRISTINA RAMIREZ: 1
31 KATHY JAMES: 1
32 THERESA WATSON: 1
33 BEVERLY BROOKS: 1
34 DENISE KELLY: 1
35 TAMMY SANDERS: 1
36 IRENE PRICE: 1
37 JANE BENNETT: 1
38 LORI WOOD: 1
39 RACHEL BARNES: 1
```

Note: this screenshot shows only the first 39 lines of 542.out. The file continues past the bottom of the screenshot, so the remaining lines are not shown.

4. Large provided test: names_long_redundant3.txt

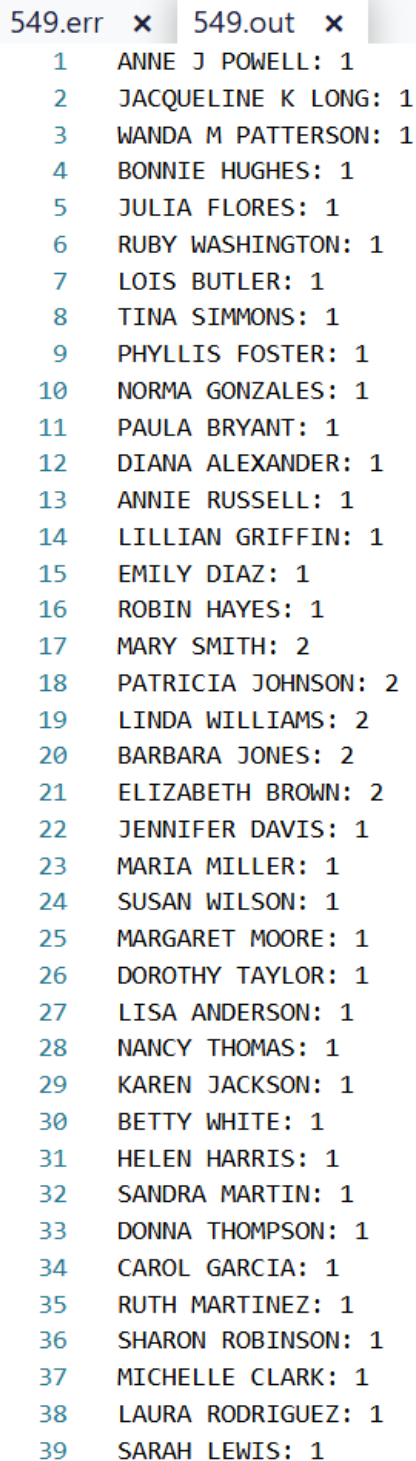
Child 549: ./shell test/names_long_redundant3.txt. The .out screenshot is truncated (first 39 lines only).

549.err (stderr)



```
549.err x 549.out x
1
```

549.out (stdout)



```
549.err x 549.out x
1 ANNE J POWELL: 1
2 JACQUELINE K LONG: 1
3 WANDA M PATTERSON: 1
4 BONNIE HUGHES: 1
5 JULIA FLORES: 1
6 RUBY WASHINGTON: 1
7 LOIS BUTLER: 1
8 TINA SIMMONS: 1
9 PHYLLIS FOSTER: 1
10 NORMA GONZALES: 1
11 PAULA BRYANT: 1
12 DIANA ALEXANDER: 1
13 ANNIE RUSSELL: 1
14 LILLIAN GRIFFIN: 1
15 EMILY DIAZ: 1
16 ROBIN HAYES: 1
17 MARY SMITH: 2
18 PATRICIA JOHNSON: 2
19 LINDA WILLIAMS: 2
20 BARBARA JONES: 2
21 ELIZABETH BROWN: 2
22 JENNIFER DAVIS: 1
23 MARIA MILLER: 1
24 SUSAN WILSON: 1
25 MARGARET MOORE: 1
26 DOROTHY TAYLOR: 1
27 LISA ANDERSON: 1
28 NANCY THOMAS: 1
29 KAREN JACKSON: 1
30 BETTY WHITE: 1
31 HELEN HARRIS: 1
32 SANDRA MARTIN: 1
33 DONNA THOMPSON: 1
34 CAROL GARCIA: 1
35 RUTH MARTINEZ: 1
36 SHARON ROBINSON: 1
37 MICHELLE CLARK: 1
38 LAURA RODRIGUEZ: 1
39 SARAH LEWIS: 1
```

Note: this screenshot shows only the first 39 lines of 549.out. The file continues past the bottom of the screenshot, so the remaining lines are not shown.

5. Multiple files: names1.txt and names2.txt

One command with two arguments; shell forks two children, one per file.

Terminal

```
→ ./shell test/names1.txt test/names2.txt
Child 430 terminated normally with exit code: 0
Child 431 terminated normally with exit code: 0
→
```

Child 430 - test/names1.txt

430.out (stdout)

```
430.err x 430.out x 431.err x 431.out x
1   Tom Wu: 3
2
```

430.err (stderr)

```
430.err x 430.out x 431.err x 431.out x
1   Warning - file test/names1.txt line 3 is empty.
2
```

Child 431 - test/names2.txt

431.out (stdout)

```
430.err x 430.out x 431.err x 431.out x
1   Jenn Xu: 2
2   Tom Wu: 1
3
```

431.err (stderr)

```
430.err x 430.out x 431.err x 431.out x
1
```

6. Same file twice: names2.txt

The same file passed twice produces two independent children with identical output.

Terminal

```
→ ./shell test/names2.txt test/names2.txt
Child 444 terminated normally with exit code: 0
Child 445 terminated normally with exit code: 0
→
```

Child 444 - test/names2.txt

444.out (stdout)

```
444.out x 444.err x 445.out x 445.err x
1 Jenn Xu: 2
2 Tom Wu: 1
3
```

444.err (stderr)

```
444.out x 444.err x 445.out x 445.err x
1
```

Child 445 - test/names2.txt

445.out (stdout)

```
444.out x 444.err x 445.out x 445.err x
1 Jenn Xu: 2
2 Tom Wu: 1
3
```

445.err (stderr)

```
444.out x 444.err x 445.out x 445.err x
1
```

7. Student test: mytestA2.txt

Our own test file passed twice; both children count Alice, Bob and Charlie correctly.

Terminal

```
→ ./shell test/mytestA2.txt test/mytestA2.txt
Child 459 terminated normally with exit code: 0
Child 460 terminated normally with exit code: 0
→
```

Child 459 - test/mytestA2.txt

459.out (stdout)

	459.err	459.out	460.err	460.out
1		Alice: 3		
2		Bob: 2		
3		Charlie: 1		
4				

459.err (stderr)

	459.err	459.out	460.err	460.out
1				

Child 460 - test/mytestA2.txt

460.out (stdout)

	459.err	459.out	460.err	460.out
1		Alice: 3		
2		Bob: 2		
3		Charlie: 1		
4				

460.err (stderr)

	459.err	459.out	460.err	460.out
1				

8. Input through stdin

With no arguments, one child reads standard input (piped from names1.txt); the warning names "stdin" as the file.

Terminal

```
→ cat test/names1.txt | ./shell
Child 474 terminated normally with exit code: 0
→
```

Child 474 - stdin (piped from test/names1.txt)

474.out (stdout)

```
474.err x 474.out x
1 Tom Wu: 3
2
```

474.err (stderr)

```
474.err x 474.out x
1 Warning - file stdin line 3 is empty.
2
```

9. Invalid file

A nonexistent input file prints an error and the child terminates with exit code 1.

Terminal

```
→ ./shell test/does_not_exist.txt  
error: cannot open file test/does_not_exist.txt  
Child 487 terminated normally with exit code: 1  
→ █
```